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Code | JupyterLab | Python [conda env:base] * | Anaconda Toolbox

```
Outliers []
```

[32]:

```python
#Probability Basics
sample={1,2,3,4,5}
event={2,4,6}
print("Sample space:",sample)
print("Event A:",event)
P_A=len(event)/len(sample)
print("\nProbability of Event A")
print("P(A)=",P_A)
```

```
Sample space: {1, 2, 3, 4, 5}
Event A: {2, 4, 6}

Probability of Event A
P(A)= 0.6
```

[37]:

```python
#Conditional Probability
event_B={4,5,6}
inter=event.intersection(event_B)
P_A_given_B=len(inter)/len(event_B)
print("\nEvent B:",event_B)
print("A n B:", inter)
print("P(A|B)=", P_A_given_B)
```

```
Event B: {4, 5, 6}
A n B: {4, 6}
P(A|B)= 0.6666666666666666
```

[41]:

```python
#Bayes' Theorm
P_Disease=0.01
P_Positive_Disease=0.99
P_Positive=0.05
P_Disease_given_Positive=(
```

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Code | JupyterLab | Python [conda env:base] * | Anaconda Toolbox

```
Outliers []
```

[32]:
```
#Probability Basics
sample-{1,2,3,4,5}
event={2,4,6}
print("Sample space:",sample)
print("Event A:",event)
P_A=len(event)/len(sample)
print("\nProbability of Event A")
print("P(A)=",P_A)
```

```
Sample space: {1, 2, 3, 4, 5}
Event A: {2, 4, 6}

Probability of Event A
P(A)= 0.6
```

[37]:
```
#Conditional Probability
event_B={4,5,6}
inter-event.intersection(event_B)
P_A_given_B=len(inter)/len(event_B)
print("\nEvent B:",event_B)
print("A n B:", inter)
print("P(A|B)-", P_A_given_B)
```

```
Event B: {4, 5, 6}
A n B: {4, 6}
P(A|B)= 0.6666666666666666
```

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P_Disease-0.01
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P_Disease=0.01
P_Positive_Disease=0.99
P_Positive=0.05
P_Disease_given_Positive=(
    P_Positive_Disease*P_Disease
)/P_Positive
print("\n Bayes' Theorm Example")
print("P(Disease)=",P_Disease)
print("P(Positive|Disease)=",P_Positive_Disease)
print("P(Positive)=",P_Positive)
print(
    "P(Disease|Positive)=",round(P_Disease_given_Positive,4)
)
print(
    "Percentage=",
    round(P_Disease_given_Positive * 100,2),"%")
```

```
 Bayes' Theorm Example
P(Disease)= 0.01
P(Positive|Disease)= 0.99
P(Positive)= 0.05
P(Disease|Positive)= 0.198
Percentage= 19.8 %
```

[45]:

```python
#Correlation
import pandas as pd
df=pd.DataFrame({
    'Hours':[1,2,3,4,5]
```

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    'Hours':[1,2,3,4,5],
    'Marks':[20,30,40,50,60]
})
corr=df['Hours'].corr(df['Marks'])
print("Pearson Correlation:",corr)
```

```
Pearson Correlation: 1.0
```

[46]:

```python
#Spearman Correlation(monotonic)
import pandas as pd
df=pd.DataFrame({
    'Hours':[1,2,3,4,5],
    'Marks':[20,30,40,50,60]
})
corr=df['Hours'].corr(
    df['Marks'],
method='spearman'
)
print("Spearman Correlation:",corr)
```

```
Spearman Correlation: 0.9999999999999999
```

[52]:

```python
#Spearman Correlation(monotonic)
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

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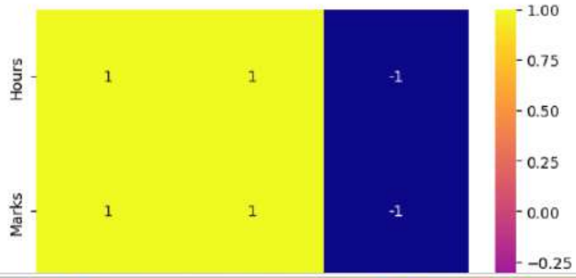
import matplotlib.pyplot as plt
df=pd.DataFrame({
    'Hours':[1,2,3,4,5],
    'Marks':[20,30,40,50,60],
    'Sleep':[9,8,7,6,5]
})
sns.heatmap(df.corr(),
annot=True,
cmap='plasma')
plt.show()
```



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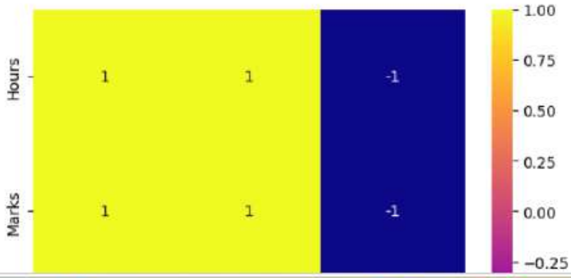

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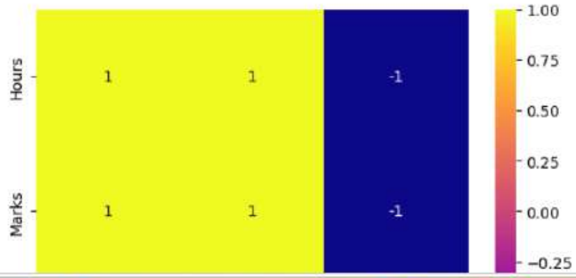

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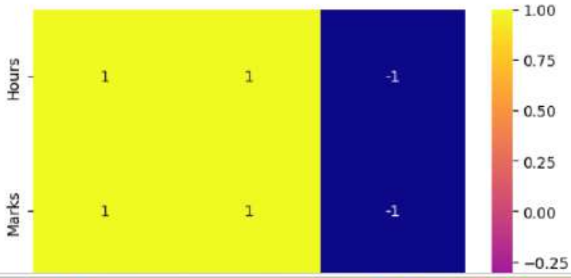

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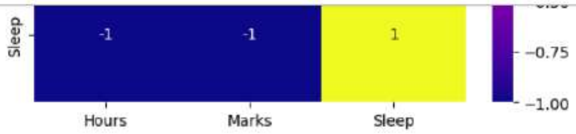

[54]:
```python
import numpy as np
import matplotlib.pyplot as plt
#Generate normally distributed data
data=np.random.normal(
    loc=70,
    scale=10,
    size=1000
)
mean=np.mean(data)
std=np.std(data)
print("Mean:",round(mean,2))
print("Standard Deviation:",round(std,2))
#Plot Histogram
plt.figure(figsize=(8,5))
sns.histplot(data,bins=30,kde=True)
plt.title("Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
```

```
Mean: 69.95
Standard Deviation: 9.64
```

[54]: `Text(0, 0.5, 'Frequency')`

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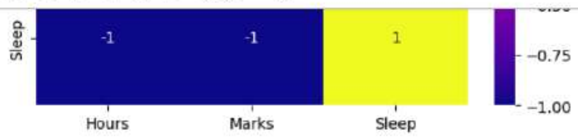
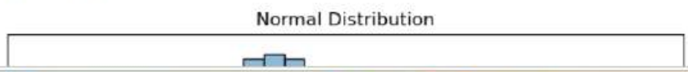


[54]:

```python
import numpy as np
import matplotlib.pyplot as plt
#Generate normally distributed data
data=np.random.normal(
    loc=70,
    scale=10,
    size=1000
)
mean=np.mean(data)
std=np.std(data)
print("Mean:",round(mean,2))
print("Standard Deviation:",round(std,2))
#Plot Histogram
plt.figure(figsize=(8,5))
sns.histplot(data,bins=30,kde=True)
plt.title("Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
```

```
Mean: 69.95
Standard Deviation: 9.64
```

[54]: `Text(0, 0.5, 'Frequency')`



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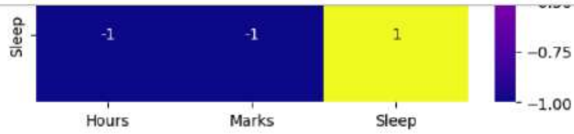

[54]:
```
import numpy as np
import matplotlib.pyplot as plt
#Generate normally distributed data
data=np.random.normal(
    loc-70,
    scale=10,
    size-1000
)
mean-np.mean(data)
std=np.std(data)
print("Mean:",round(mean,2))
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plt.figure(figsize=(8,5))
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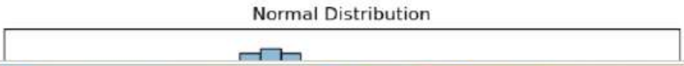

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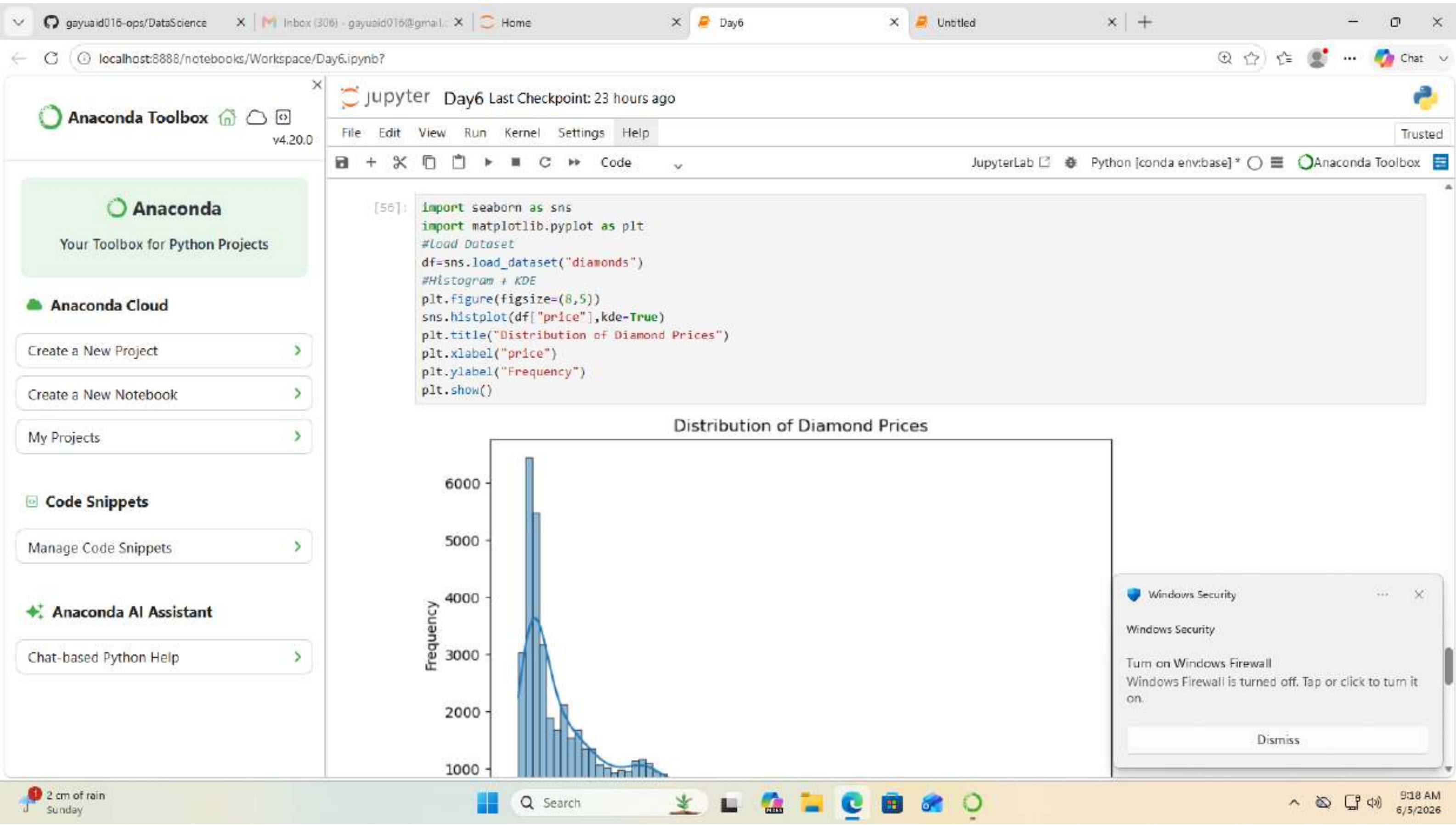
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[56]:
import seaborn as sns
import matplotlib.pyplot as plt
#Load Dataset
df=sns.load_dataset("diamonds")
#Histogram + KDE
plt.figure(figsize=(8,5))
sns.histplot(df["price"],kde=True)
plt.title("Distribution of Diamond Prices")
plt.xlabel("price")
plt.ylabel("Frequency")
plt.show()
Distribution of Diamond Prices
6000
5000
4000
3000
2000
1000
Frequency
Windows Security
Windows Security
Turn on Windows Firewall
Windows Firewall is turned off. Tap or click to turn it on.
Dismiss
2 cm of rain
Sunday
Search
9:18 AM
6/5/2026

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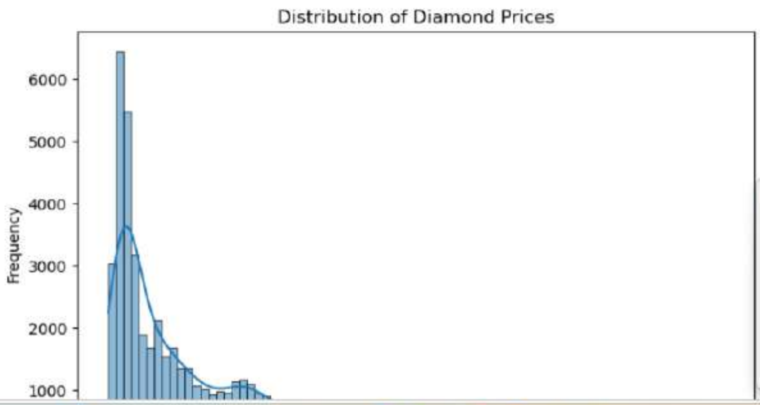

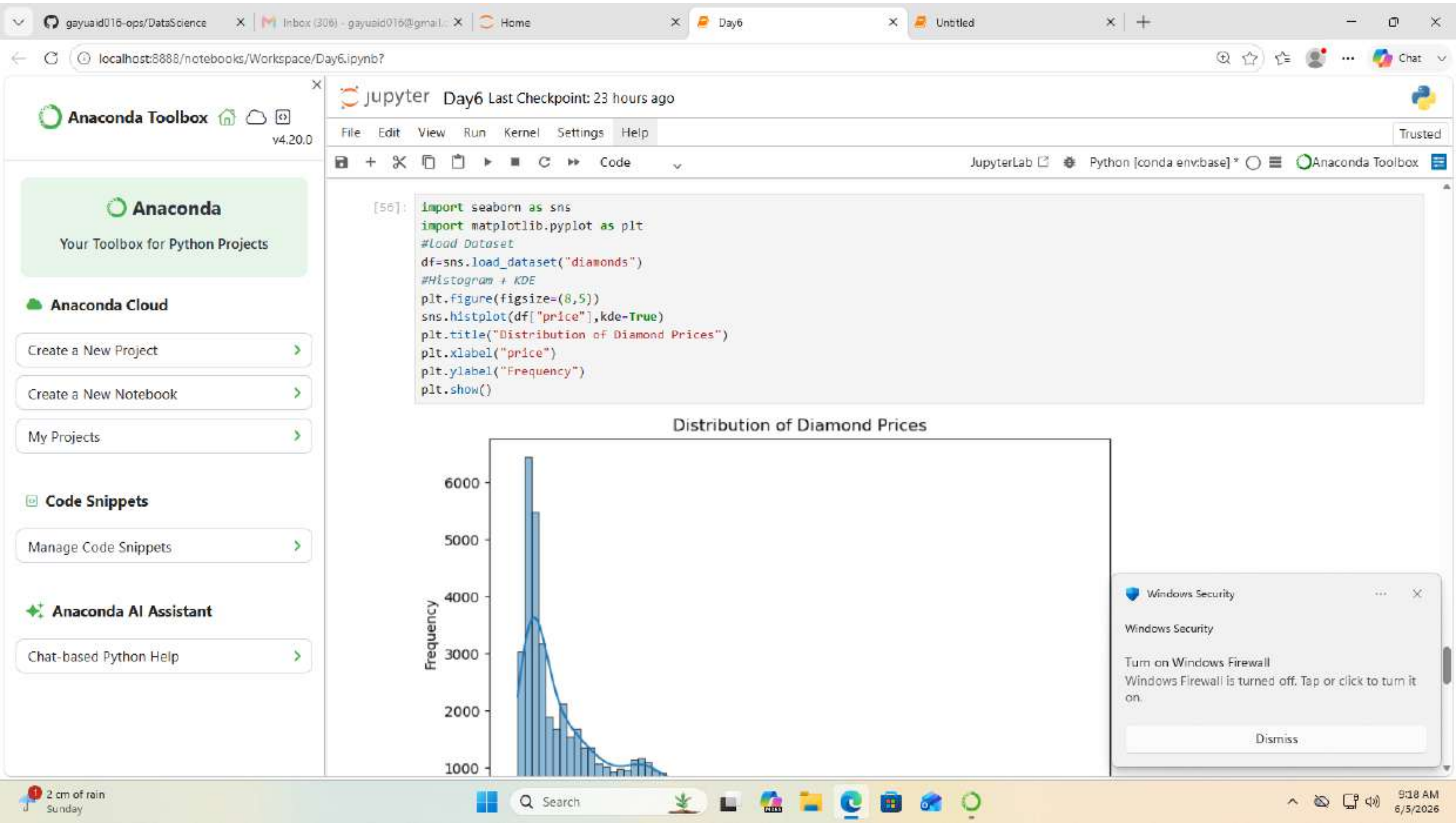

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